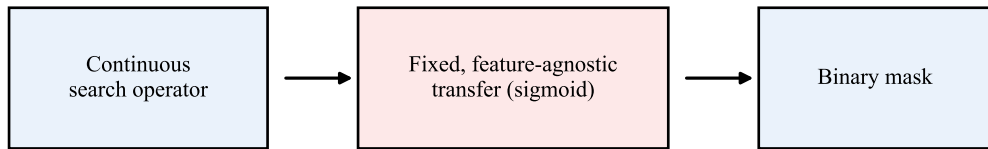
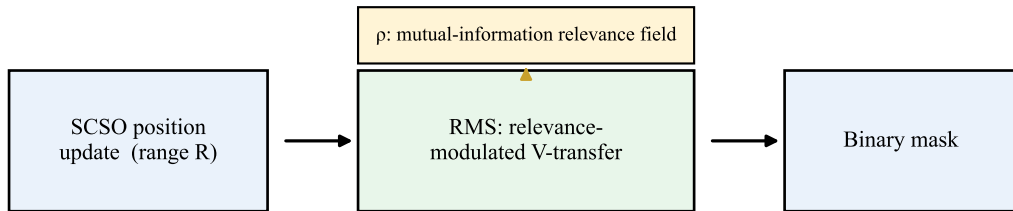


**(a) Conventional continuous-to-binary pipeline**



*quantization discards fine adjustments → WASHOUT*

**(b) RG-SCSO: sensitivity range → per-feature bit-flip**



*UMR: memetic refinement on uncertain bits*